

lines making cars, but none of them resemble our first idea of a robot. There are humanoid robots out there, but it is still a developing field with many challenges. Let's take a look at the current generation of humanoids and see why they are nowhere close to what we see in fiction.

Current generation of humanoid robots and their capabilities

A humanoid robot is a robot designed to resemble the human body and has features and functions that effectively resemble ours. There are a number of reasons why humanoids can blend in more easily and are more desirable, especially when considering home settings where they are surrounded by humans. This is not only because they look and feel like humans making them look more familiar and trustworthy, but also because they can interact with various human tools and environments more easily.

There are a number of reasons why Humanoids are a very interesting field for robotics engineer and scientists. Humanoids need researchers to study and understand the details of how exactly the human body works and use the knowledge to make better and stronger humanoid robots.

The focus earlier used to be on using this research to develop better prosthetics for humans but the two fields closely overlap each other and the knowledge has been transferred between the two disciplines leading to faster development of both advanced prosthetics and humanoid robots.

Humanoids are suitable for a number of tasks, specially various desk and clerical job roles that don't require many specific skill sets. The focus of humanoid development has mainly been on developing robots that look, walk, talk and act like humans so they can take up various roles such as become companions for the sick and elderly, sit as receptionists or other similar desk jobs, and take place of a human in dangerous situations. The main advantage of humanoids is that, in theory, since they look like humans and their body is designed like ours, they can take up any task that a human can do, albeit they have been programmed to do so. In theory, a humanoid can be capable enough to replace humans in almost anything they do.

But 'In theory' doesn't translate to 'In reality' every time. Look around you – how many robot receptionists or robot firefighters do you see? The current generation of humanoid robots is nowhere close to replacing all of us everywhere and there are a number of reasons and challenges as to why. Let's take a look at some of the humanoids of the current generation and explore some of these reasons before we look at possible ways to overcome these challenges to making them ubiquitous.